Multivariable Calculus Final Project

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1 Construction

1.1 Introduction & Design

For this project, our group chose a cone with radius 30" and height 36" (or $2.5'$ and $3'$, respectively). The equation that we used was

$$\frac{x^2}{225} + \frac{y^2}{225} = \frac{z^2}{900},$$

where $x \in [-15, 15]$, $y \in [-15, 15]$, and $z \in [0, 30]$. Our cone was domain-restricted to only positive z values, as it would be physically very difficult to construct a cone that extended into the ground.

1.2 Construction Process

Our construction got off to a rough start, with our group not acquiring the proper materials until . . . .

2 Calculations

2.1 Theoretical Precise Value

To calculate the theoretical precise value, we will use the triple integral

$$\iiint_Q f(x, y, z)\, dV.$$

To find the bounds, we will use cylindrical coordinates, as Cartesian coordinates tend to end up messy in such calculations. To find the bounds in cylindrical coordinates, we only need to solve for r, as $\theta \in [0, 2\pi]$ and $z \in [0, 30]$, as given by our domain restrictions above.

To find the bounds for r, we first convert the rectangular x and y to cylindrical coordiantes. Following the conversion, our cone can be represented by the equation $\frac{r^2}{225} = \frac{z^2}{900}$, which we can solve to find the upper bound for r:

$$\frac{r^2}{225} = \frac{z^2}{900} \implies r^2 = \frac{225z^2}{900} \implies r = \sqrt{\frac{225z^2}{900}} = \sqrt{\frac{z^2}{4}} = \frac{z}{2}.$$

Thus, the upper bound for r is $\frac{z}{4}$, and because the lower bound for r is 0, we know that the complete bounds for r are $[0, \frac{z}{4}]$. Thus, the complete bounds, for all variables, are:

$$\theta \in [0, 2\pi]\,,\ z \in [0, 30]\,,\ \text{and } r \in \left[0, \frac{z}{2}\right].$$

Therefore, we can plug in the bounds and integrand for the triple integral and evaluate it:

$$V = \int_0^{2\pi} \int_0^{30} \int_0^{\frac{z}{2}} r\, dr\, dz\, d\theta = \int_0^{2\pi} \int_0^{30} \left[\frac{r^2}{2}\right]_0^{\frac{z}{2}} dz\, d\theta = \int_0^{2\pi} \int_0^{30} \frac{z^2}{8}\, dz\, d\theta$$
$$= \int_0^{2\pi} \left[\frac{z^3}{24}\right]_0^{30} d\theta = 1125 \int_0^{2\pi} d\theta = 1125\,(2\pi) = 2250\pi \approx 7068.5835.$$

Thus, the theoretical precise volume of our cone is approximately $\boxed{7068.5835 \text{ cubic inches}}$ (or approxinately 4.0906 cubic feet).

2.2 Theoretical Approximate Value

To calculate the theoretical approximate value of the equation $\frac{x^2}{225} + \frac{y^2}{225} = (\frac{z^2}{900})$, we must first convert to cylindrical coordinates to find the radius with respect to z, the height. The radius r can be found via the equations:

$$\frac{r^2}{225} = \frac{z^2}{900} \implies r^2 = \frac{z^2}{4} \implies r = \frac{z}{2}.$$

To visualize the Riemann sum, imagine a layer cake with multiple layers that are cylinders of arbitrary h height, and gradually increasing radii. A summation of all the heights of every layer equates to z, or 30. And, note that there are n cylinders. Additionally, at the peak of the inverted cone, or height 30, the radius is 15, and at the minimum, or height 1 (not including height 0), the radius is $\frac{1}{2}$. Now, for the formal calculation.

Let r be the radius of the partition which is a cylinder, h be the height, z be the height of the original cone, and n be the number of partitions that are taken, specifically referring to the height. The radius, with respect to z, is $r = \frac{z}{2}$. The area of each cylinder's circle is $\frac{z^2}{4}\pi$. Therefore, the area of each partition is $\frac{z^2}{4}h\pi$. Finally, let $\Delta z_0 = \frac{30-0}{n}$, and let $z_i = i\Delta z_0$. Inputting all the values into the Riemann sum and taking the limit of it is as follows:

$$V = \sum_{i=1}^{n}[f(z_i)]^2\pi\Delta z_0 = \sum_{i=1}^{n}\frac{z_i^2}{4}\pi\Delta z_0 = \sum_{i=1}^{n}\frac{(i\Delta z_0)^2\pi}{4}\Delta z_0 = \sum_{i=1}^{n}\frac{\pi}{4}i^2\Delta z_0^3 = \sum_{i=1}^{n}\frac{\pi}{4}i^2\frac{30^3}{n^3}$$
$$= \sum_{i=1}^{n}\frac{\pi}{4}i^2\frac{27000}{n^3} = \lim_{n\to\infty}\sum_{i=1}^{n}\frac{6750\pi}{n^3}i^2.$$

Now, we hit a roadblock, as the value of $\sum_{i=1}^{n} i^2$ is unknown. Since i is the variable that changes each partition to the next, the summation is equal to $1^2 + 2^2 + 3^2 + \cdots + n^2$, and the explicit formula is $\frac{n(n+1)(2n+1)}{6}$, via the closed-form identity. Thus, the final Riemann sum is

$$V = \lim_{n\to\infty}\sum_{i=1}^{n}\frac{6750\pi}{n^3}\left[\frac{n(n+1)(2n+1)}{6}\right] = \frac{6750\pi}{3} \approx 7068.5835.$$

Thus, the theoretical approximate value of our cone is approximately $\boxed{\text{7068.5835 cubic inches}}$ (or approximately 4.09986 cubic feet.)

3 Discussion

3.1 Theoretical Precise Value

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3.2 Theoretical Approximate Value

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